Shiva Sharma

R&D Software Engineer · C / C++ · Embedded Linux · Optical Networking

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Portfolio · GitHub · LinkedIn

Professional summary

Embedded software engineer with 3+ years of hands-on C/C++ development experience in carrier-class optical networking and telecom systems on Linux. Proven ability to deliver production-quality features end-to-end—from system design through integration, testing, and release—within cross-functional agile teams. Deep domain expertise in OTN control plane, GPON/XGSPON/NGPON, OMCI, QoS, GCC, and inter-chassis networking, with a strong focus on code quality and automated testing.

Work experience

Engineer | Tejas Networks
May 2025 – Present

- Designing and implementing C/C++ control plane software for Optical Transport Network (OTN) devices in a Linux embedded environment, improving network reliability and operational efficiency.
- Developed and optimized Generic Control Channel (GCC) handling for Line and Client Ports, enabling reliable embedded management communication across OTN links.
- Implemented inter-card and inter-chassis connectivity solutions, enabling seamless communication across distributed hardware in multi-card systems.
- Developing and enhancing Wi-Fi 6 ONT Mesh technology, optimizing mesh networking performance across the product portfolio.

R&D; Engineer | Nokia
Aug 2023 – Apr 2025

- Developed embedded C/C++ software for SFU ONT devices, specializing in QoS data packet management and OMCI-level communication across GPON, XGSPON, and NGPON.
- Delivered major new features end-to-end—from design and implementation to customer release—enhancing product functionality and maintainability.
- Led design pattern and technology selection for new modules while navigating chipset compatibility, vendor limitations, and integration with existing systems in a multi-threaded Linux environment.
- Developed scalable, maintainable solutions under real-world constraints from multiple chipsets and third-party modules.
- Conducted rigorous code reviews and maintained automated test suites, upholding high standards of code quality and system stability.
- Collaborated closely with hardware engineers, software teams, and third-party vendors to ensure seamless system integration.

Student Trainee (Intern) | Nokia Solutions and Networks
10 months

- Worked on the L2 forwarding team for SFU ONTs, developing and maintaining embedded software for Optical Network Terminal devices.
- Enhanced ONT device performance through continuous software updates, debugging, and troubleshooting in a Linux environment.
- Received a letter of appreciation; performance led to conversion to a full-time R&D; Engineer role.

Selected projects

OTN Control Plane Simulator | C / Linux
A distributed control-plane simulator using epoll, UDP discovery, heartbeats, and reliable TCP state synchronization. Demonstrates systems programming, network protocols, concurrency, and fault-aware communication.

Gatekeeper OpenWrt | Dart / Networking
A networking-focused project exploring software interfaces for connected OpenWrt devices.

Shiksha Anudan | Flutter / Mobile
A mobile application project focused on making education support more accessible.

Technical skills

Programming languages	C · C++ (advanced / multi-threaded) · SQL
Embedded & systems	Embedded Linux · Multi-threaded systems · Network protocols · Inter-process communication · Debugging
Networking domains	OTN control plane · GCC (Line & Client Ports) · GPON / XGSPON / NGPON · OMCI · QoS · Inter-card / inter-chassis networking · Wi-Fi 6 Mesh
Tools & practices	Linux · Agile / TDD · CI/CD pipelines · Code review · Automated testing · Git

Education

Master of Computer Applications — 2023
Vellore Institute of Technology

Bachelor of Computer Applications — 2021
Gujarat University / St. Xavier's College

Class 12 — 2018
Birla International School

Certifications

SQL — HackerRank · C++ — Coding Blocks

Additional information

Portfolio: shivasharma9896.github.io/portfolio-site
GitHub: github.com/shivasharma9896